Souptik Majumdar

Stuttgart, Germany | +49 17685984542 | souptikmajumdar98@gmail.com | souptik.majumdar@vis.uni-stuttgart.de |
https://www.linkedin.com/in/souptikmajumdar/

Summary

Machine Learning PhD researcher with 5+ years of prior industry experience in cloud, automation, and data engineering (Bosch, Oracle, Deloitte). Expertise in multimodal LLMs, spatial vision, and world models. Actively targeting applied industry roles to focus on shipping scalable, high-impact AI products and solving real-world engineering challenges.

Education and Achievements

University of Stuttgart Stuttgart, BW, Germany
PhD in Computer Science, Collaborative AI (CAI) Group – Prof. Andreas Bulling *Apr. 2025 – Present*

- Thesis: **"Understanding Minds Through Collaboration"** – egocentric collaborative Theory of Mind in vision-language models.
- Member of the **International Max Planck Research School for Intelligent Systems (IMPRS-IS)**.

University of Stuttgart Stuttgart, BW, Germany
M.Sc. Computer Science, ***GPA 1.5*** *Oct. 2022 – Mar. 2025*

- Master thesis at Bosch on efficient vision models (Mixture of Experts, depth-adaptive computation, knowledge distillation).
- **Deutschlandstipendium** scholarship (2024 – 2025).

University of Calcutta Kolkata, WB, India
B.Sc. Computer Science, ***GPA 1.4*** *(German-equivalent)* *Jun. 2016 – Jun. 2019*

Research Experience

Doctoral Researcher – Collaborative AI Group Apr. 2025 – Present
University of Stuttgart *Stuttgart, BW, Germany*

- Thesis: *Understanding Minds Through Collaboration* studies **egocentric collaborative Theory of Mind in vision-language models** – when a model watching a task from a first-person view can infer what its partner sees, knows, and intends, and act on it.
- Combines **benchmark construction**, **mechanistic interpretability** and **activation steering** of multimodal LLMs, and **cooperative multi-agent RL** with latent world models to connect what a model represents about others' minds to how it acts with them.
- Supervising Master's and course projects on gaze-guided efficient VLMs and video Theory of Mind;
- **Gamescom 2026** as part of the **PRETINA project** (funded by Bundesministerium für Forschung, Technologie und Raumfahrt): demonstrated **AbracaData**, a VR game exploring privacy-preserving eye tracking in gaming, and discussed how gaze data can inform emotion, health, and player experience.

Research Intern and Master Thesis – Efficient Deep Learning, Computer Vision June 2024 – Mar. 2025
Bosch GmbH *Renningen, BW, Germany*

- Made vision models efficient with **Mixture of Experts** (MoE) and **Mixture-of-Depths** routing, reducing **FLOPs** and inference time of large vision and language models.
- Applied **Knowledge Distillation** to transfer knowledge from large models to smaller ones without significant loss in accuracy; work published at an **ICLR 2025 Workshop**.

Tutorial Assistant – Reinforcement Learning; Artificial Intelligence Oct. 2023 – Aug. 2024
Institute of AI, University of Stuttgart *Stuttgart, BW, Germany*

- Led tutorial sessions on RL (value/policy iteration, deep RL) and on foundations of AI and autonomous systems.

Industry Experience

ECU Data Analyst Mar. 2023 – May 2024
Daimler Truck AG *Stuttgart, BW, Germany*

- Built **ETL** pipelines with Jupyter and **Azure Data Factory** to visualize vehicle CAN-bus data; automated **ECU** software updates in Python/C++; engineered predictive-diagnostics applications.

Cloud AI Integration Developer Dec. 2021 – Sep. 2022
Oracle *Hyderabad, India*

- Designed and deployed Oracle Cloud integrations with AI-based endpoints between Oracle Cloud SaaS and on-premise systems; built **REST APIs** with Python/Flask and Oracle REST Data Services.

Cloud & Robotic Process Automation Developer Aug. 2019 – Dec. 2021
Deloitte US *Hyderabad, India*

- Designed **automation bots** (UiPath, Automation Anywhere, Bash) for Oracle Cloud financial period-close processes, now used in over 100 countries; automated BI reporting; managed **CI/CD pipelines** in Oracle VBCS.
- Built proofs of concept with **Google Cloud Document AI** (invoice processing) and Google Cloud AI APIs (automatic email translation).

Publications and Preprints

CARD: Mechanistic interpretability of belief-to-action routing in VLMs	Under review, EMNLP 2026
DreamTeam: Cooperative multi-agent RL with latent world models (second author)	Under review, NeurIPS 2026
Mixture-of-Depths Routing for Efficient Vision Models	ICLR 2025 **SCOPE**
Using Deep Learning to Recognize Emotions Through Speech Analysis (Springer)	Sep. 2023
Route Optimization Application using Server-Client Architecture and Google APIs (IEEE)	Feb. 2020

Technical Skills & Certifications

Research Areas: Theory of Mind, Vision-Language Models, Mechanistic Interpretability, Activation Steering, Multi-Agent & Cooperative RL, World Models, Efficient Deep Learning, Egocentric Vision, Gaze/Eye Tracking
Deep/Machine Learning: PyTorch, JAX (JAXMARL), Transformers, LLMs/VLMs, RAG, Knowledge Distillation, MoE, GRPO, TensorFlow, Keras, Wandb, MLflow, Pandas, NumPy, Matplotlib
Programming: Python, C/C++, Java, JavaScript, HTML/CSS, SQL/PLSQL, Bash, LaTeX
Cloud & Tools: Git, Linux GPU clusters, Azure Data Factory, GCP, AWS, Oracle Cloud / Integration Cloud, Automation Anywhere, UiPath, VS Code
Certifications: Oracle Cloud Platform Application Integration 2021 Certified Specialist; Complete Data Science Bootcamp (Udemy)

Selected Projects

Grammar Variational Auto-Encoders for Transparent Deep Learning | *PyTorch, Generative AI, Explainable AI*

- Built AE, VAE, and **Grammar VAE** models to encode arithmetic equations; grammar-rule encoding improved accuracy and produced semantically correct equations versus character-level baselines.

Joint Gaze–Head Representation Learning for Data Augmentation | *PyTorch, Multi-Modal Transformers*

- Used **multi-modal Transformers** to reconstruct missing gaze data from IMU head-movement data; outperformed interpolation baselines on downstream gaze-estimation tasks.

COVID-19 Data Analysis and Reporting on Azure and Oracle Cloud | *Azure Data Factory, ETL, BI*

- End-to-end pipelines ingesting HTTP data into Azure Data Lake; interactive BI reports via Oracle Integration Cloud and SQL/PLSQL with scheduled triggers.

Languages & Visa

English: C1 | **German:** B1
Work Authorization: EU Blue Card holder (Germany) – no sponsorship required